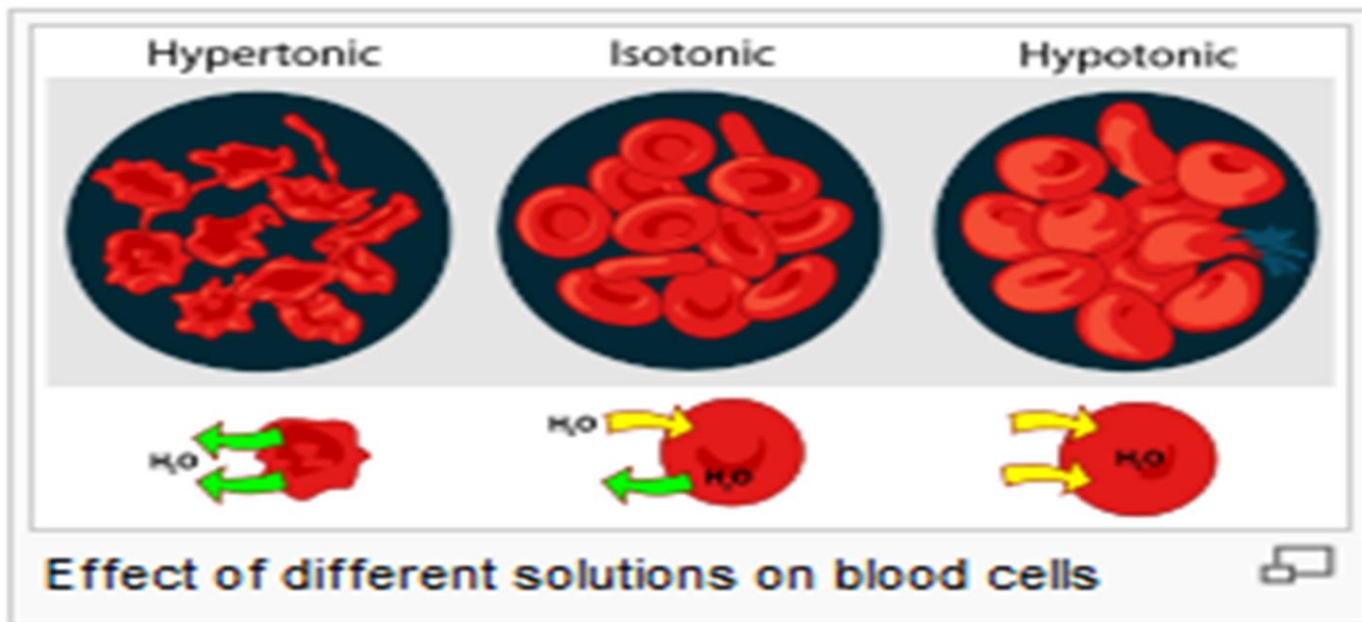


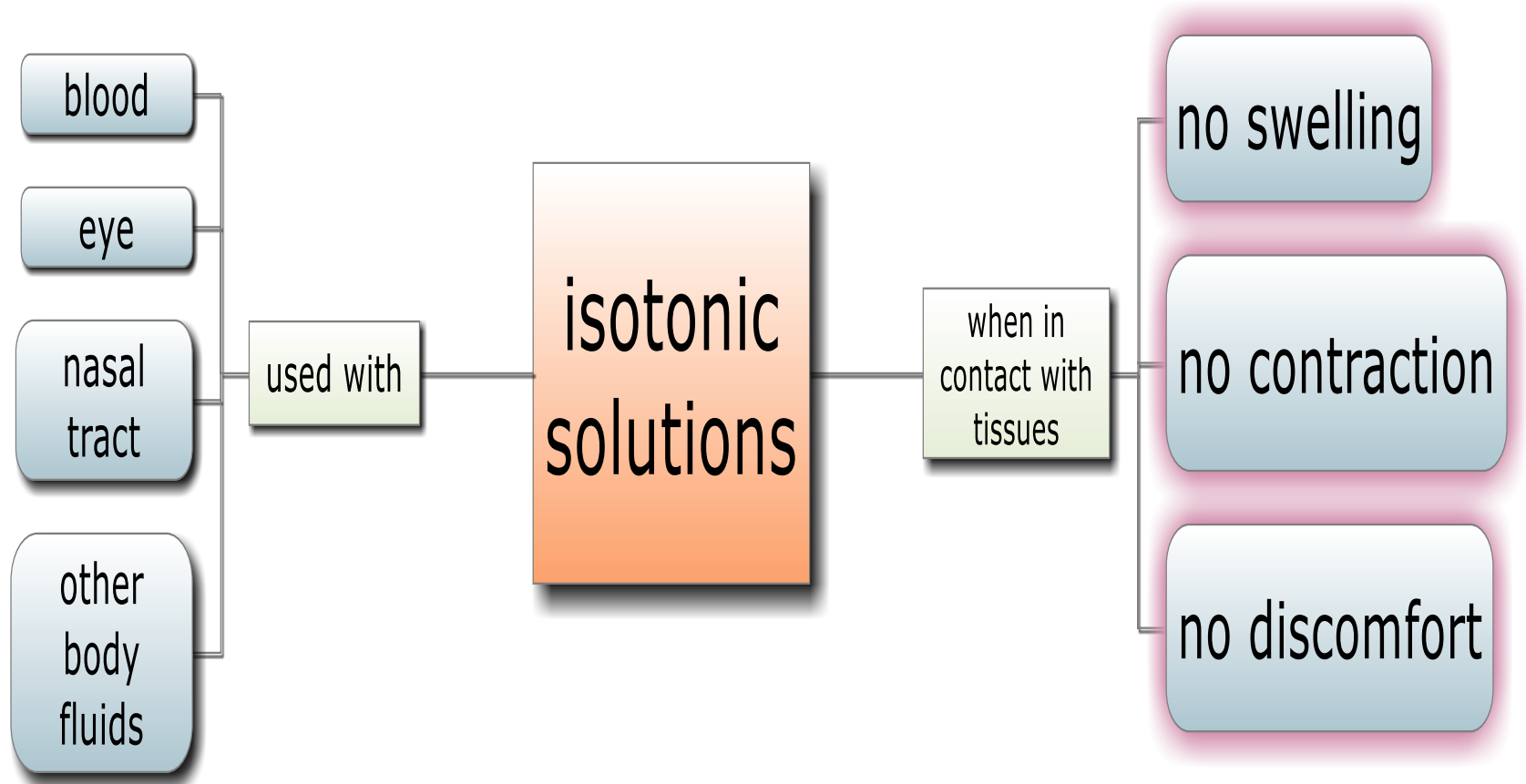
Isotonic & Buffer Solutions



Objectives

- Upon successful completion of this chapter, you will be able to:
 - ✓ Differentiate between the terms *isosmotic*, *isotonic*, *hypertonic*, and *hypotonic*.
 - ✓ Apply physical chemical principles in the calculation of isotonic solutions.
 - ✓ Perform the calculations required to prepare isotonic compounded prescriptions.
 - ✓ State the buffer equation and apply it in calculations.

Why using isotonic solutions?



Isotonic

- When a solvent passes through a semipermeable membrane from a dilute solution into a more concentrated one, the concentrations become equalized and the phenomenon is known as ***osmosis***.
- The pressure responsible for this phenomenon is termed ***osmotic pressure*** and varies with the nature of the solute.
- If the solute is a **nonelectrolyte**, its solution contains only molecules and the osmotic pressure varies with the concentration of the solute.

Isotonic Cont'd

- If the solute is an **electrolyte**, its solution contains ions and the osmotic pressure varies with both the concentration of the solute and its degree of dissociation.
- Thus, solutes that dissociate present a greater number of particles in solution and exert a greater osmotic pressure than *undissociated* molecules.
- Like osmotic pressure, the other ***colligative properties*** of solutions, *vapor pressure*, *boiling point*, and ***freezing point***, depend on the number of particles in solution.

Isotonic Cont'd

- Therefore, these properties are interrelated and a change in any one of them will result in a corresponding change in the others.
- Two solutions that have the same osmotic pressure are termed ***isosmotic***.
- Many solutions intended to be mixed with body fluids are designed to have the same osmotic pressure for greater **patient comfort, efficacy, and safety**.
- A solution having the same osmotic pressure as a *specific* body fluid is termed ***isotonic*** (meaning of equal tone) with *that* specific body fluid.

Isotonic Cont'd

- Solutions of *lower* osmotic pressure than that of a body fluid are termed ***hypotonic***, whereas those having a *higher* osmotic pressure are termed ***hypertonic***.
- **Pharmaceutical dosage forms** intended to be added directly to the blood or mixed with biological fluids of the eye, nose, and bowel are of principal concern to the pharmacist in their preparation and clinical application.

Special Clinical Consideration of Tonicity

- It is generally accepted that for ophthalmic and parenteral administration, isotonic solutions are better tolerated by the patient than those at the **extremes of hypo- and hyper-tonicity**.
- With the administration of an isotonic solution, there is a **homeostasis** with the body's intracellular fluids.
- Thus, *in most instances*, preparations that are isotonic, or nearly so, are preferred.
- However, there are exceptions, as in instances in which hypertonic solutions are used to “draw” fluids out of edematous tissues and into the administered solution.

Tonicity Cont'd

- Most ophthalmic preparations are formulated to be isotonic, or approximately isotonic, to duplicate ophthalmic tears for the comfort of the patient.
- These solutions are also prepared and buffered at an appropriate pH, both to reduce the likelihood of irritation to the eye's tissues and to maintain the stability of the preparations.
- Injections that are not isotonic should be administered slowly and in small quantities to **minimize tissue irritation, pain, and cell fluid imbalance.**

Tonicity Cont'd

- The tonicity of small-volume injections is generally inconsequential when added to large-volume parenteral infusions because of the presence of tonic substances, such as sodium chloride or dextrose in the large-volume infusion, which **serve to adjust the tonicity** of the smaller added volume.
- Intravenous infusions, which are hypotonic or hypertonic, can have profound adverse effects because they generally are administered in large volumes.
- Large volumes of *hypertonic* infusions containing dextrose, for example, can result in **hyperglycemia, osmotic diuresis, and excessive loss of electrolytes.**

Tonicity Cont'd

- Excess infusions of *hypotonic* fluids can result in the osmotic **hemolysis** of red blood cells and surpass the upper limits of the body's capacity to safely absorb excessive fluids.
- Even isotonic fluids, when infused intravenously in excessive volumes or at excessive rates, can be **deleterious** due to an overload of fluids placed into the body's circulatory system.

Physicochemical Considerations in the preparation of Isotonic Solutions

- The calculations involved in preparing isotonic solutions may be made in terms of data relating to the **colligative** properties of solutions.
- Theoretically, any one of these properties may be used as a basis for determining tonicity.
- Practically and most conveniently, a comparison of freezing points is used for this purpose.
- It is generally accepted that **0.52 C is the freezing point** of both blood serum and lacrimal fluid.

Isotonic Preparation Cont'd

- When one gram molecular weight of any nonelectrolyte, that is, a substance with negligible dissociation, such as boric acid, is dissolved in 1000 g of water, the freezing point of the solution is about **1.86C below the freezing point of pure water.**
- By simple proportion, therefore, we can calculate the weight of any nonelectrolyte that should be dissolved in each 1000 g of water if the solution is to be isotonic with body fluids.

Isotonic Preparation Cont'd

- Boric acid, for example, has a molecular weight of 61.8; thus (in theory), 61.8 g in 1000 g of water should produce a freezing point of 1.86C. Therefore:

$$\frac{1.86 (^{\circ}\text{C})}{0.52 (^{\circ}\text{C})} = \frac{61.8 (\text{g})}{x (\text{g})}$$
$$x = 17.3 \text{ g}$$

Isotonic Preparation Cont'd

- In short, 17.3 g of boric acid in 1000 g of water, having a weight-in-volume strength of approximately 1.73%, should make a solution isotonic with lacrimal fluid.
- With electrolytes, the problem is not so simple.
- Because osmotic pressure depends more on the number than on the kind of particles, substances that dissociate have a tonic effect that increases with the degree of dissociation; the greater the dissociation, the smaller the quantity required to produce any given osmotic pressure.

Isotonic Preparation Cont'd

- If we assume that sodium chloride in weak solutions is about 80% dissociated, then each 100 molecules yields 180 particles, or 1.8 times as many particles as are yielded by 100 molecules of a nonelectrolyte.
- This dissociation factor, commonly symbolized by the letter i , must be included in the proportion when we seek to determine the strength of an isotonic solution of sodium chloride (m.w. 58.5):

Isotonic Preparation Cont'd

$$\frac{1.86\text{ (}^{\circ}\text{C)} \times 1.8}{0.52\text{ (}^{\circ}\text{C)}} = \frac{58.5\text{ (g)}}{x\text{ (g)}}$$

$$x = 9.09\text{ g}$$

Isotonic Preparation Cont'd

- Hence, 9.09 g of sodium chloride in 1000 g of water should make a solution isotonic with blood or lacrimal fluid.
- In practice, a 0.90% w/v sodium chloride solution is considered isotonic with body fluids.
 - Simple isotonic solutions may then be calculated by using this formula:

$$\frac{0.52 \times \text{molecular weight}}{1.86 \times \text{dissociation (i)}} = \text{g of solute per 1000 g of water}$$

Isotonic Preparation Cont'd

- The value of i for many a medicinal salt has not been experimentally determined.
- Some salts (such as zinc sulfate, with only some 40% dissociation and an i value therefore of 1.4) are exceptional, but most medicinal salts approximate the dissociation of sodium chloride in weak solutions.
- If the number of ions is known, we may use the following values, lacking better information.

Isotonic Preparation Cont'd

- ✓ Nonelectrolytes and substances of slight dissociation: **1.0**
- ✓ Substances that dissociate into 2 ions: **1.8**
- ✓ Substances that dissociate into 3 ions: **2.6**
- ✓ Substances that dissociate into 4 ions: **3.4**
- ✓ Substances that dissociate into 5 ions: **4.2**

Isotonic Preparation Cont'd

- A special problem arises when a prescription directs us to make a solution isotonic by adding the proper amount of some substance other than the active ingredient or ingredients.
- **Given a 0.5% w/v** solution of sodium chloride, we may easily calculate that $0.9\text{ g} - 0.5\text{ g} = 0.4\text{ g}$ of additional sodium chloride that should be contained in each 100 mL if the solution is to be made isotonic with a body fluid.
- But how much sodium chloride should be used in preparing 100 mL of a 1% w/v solution of atropine sulfate, which is to be made isotonic with lacrimal fluid? The answer depends on *how much sodium chloride is in effect represented by the atropine sulfate*.

Isotonic Preparation Cont'd

- The relative tonic effect of two substances-that is, the quantity of one that is the equivalent in tonic effects to a given quantity of the other-may be calculated if the quantity of one having a certain effect in a specified quantity of solvent is divided by the quantity of the other having the same effect in the same quantity of solvent.
- For example, we calculated that 17.3 g of boric acid per 1000 g of water and 9.09 g of sodium chloride per 1000 g of water are both instrumental in making an aqueous solution **isotonic with lacrimal fluid**.

Isotonic Preparation Cont'd

- If, however, 17.3 g of boric acid are equivalent in tonicity to 9.09 g of sodium chloride, then 1 g of boric acid must be the equivalent of $9.09 \text{ g} / 17.3 \text{ g}$ or 0.52 g of sodium chloride. Similarly, 1 g of sodium chloride must be the “tonicic equivalent” of $17.3 \text{ g} / 9.09 \text{ g}$ or 1.90 g of boric acid.

Isotonic Preparation Cont'd

- We have seen that one quantity of any substance should in theory have a constant tonic effect if dissolved in 1000 g of water: 1 g molecular weight of the substance divided by its *i* or dissociation value.
- Hence, the relative quantity of sodium chloride that is the tonic equivalent of a quantity of boric acid may be calculated by these ratios:

$$\frac{58.5 \div 1.8}{61.8 \div 1.0} \text{ or } \frac{58.5 \times 1.0}{61.8 \times 1.8}$$

Isotonic Preparation Cont'd

- And we can formulate a convenient rule: *quantities of two substances that are tonicic equivalents are proportional to the **molecular weights of each multiplied by the i value of the other.***
- To return to the problem involving 1 g of atropine sulfate in 100 mL of solution:

Molecular weight of sodium chloride = 58.5; $i = 1.8$

Molecular weight of atropine sulfate = 695; $i = 2.6$

$$\frac{695 \times 1.8}{58.5 \times 2.6} = \frac{1 \text{ (g)}}{x \text{ (g)}}$$

$x = 0.12$ g of sodium chloride represented by
1 g of atropine sulfate

Isotonic Preparation Cont'd

- Because a solution isotonic with lacrimal fluid should contain the equivalent of 0.90 g of sodium chloride in each 100 mL of solution, the difference to be added must be $0.90\text{ g} - 0.12\text{ g} = 0.78\text{ g}$ of sodium chloride.
- *If the number of grams of a substance included in a prescription is multiplied by its sodium chloride equivalent, the amount of sodium chloride represented by that substance is determined.*

The procedure for the *calculation of isotonic solutions with sodium chloride equivalents* may be outlined as follows:

Step 1. Calculate the amount (in grams) of sodium chloride represented by the ingredients in the prescription. Multiply the amount (in grams) of each substance by its sodium chloride equivalent.

Step 2. Calculate the amount (in grams) of sodium chloride, alone, that would be contained in an isotonic solution of the volume specified in the prescription, namely, *the amount of sodium chloride in a 0.9% solution of the specified volume*. (Such a solution would contain 0.009 g/mL.)

Step 3. Subtract the amount of sodium chloride represented by the ingredients in the prescription (Step 1) from the amount of sodium chloride, alone, that would be represented in the specific volume of an isotonic solution (Step 2). The answer represents the amount (in grams) of sodium chloride to be added to make the solution isotonic.

Step 4. If an agent other than sodium chloride, such as boric acid, dextrose, or potassium nitrate, is to be used to make a solution isotonic, divide the amount of sodium chloride (Step 3) by the sodium chloride equivalent of the other substance.

Example Calculations of the i Factor

Example-1.

- *Zinc sulfate is a 2-ion electrolyte, dissociating 40% in a certain concentration. Calculate its dissociation (i) factor.*
 - On the basis of 40% dissociation, 100 particles of zinc sulfate will yield:
 - 40 zinc ions
 - 40 sulfate ions
 - 60 undissociated particles
 - or 140 particles
 - Because 140 particles represent 1.4 times as many particles as were present before dissociation, the **dissociation (i) factor is 1.4 .**

Example-2.

- *Zinc chloride is a 3-ion electrolyte, dissociating 80% in a certain concentration. Calculate its dissociation (i) factor.*
 - On the basis of 80% dissociation, 100 particles of zinc chloride will yield:
 - 80 zinc ions
 - 80 chloride ions
 - 80 chloride ions
 - 20 undissociated particles
 - or 260 particles
 - Because 260 particles represents 2.6 times as many particles as were present before dissociation, the dissociation (*i*) factor is **2.6**.

Sodium Chloride Equivalents (E-Values)

SUBSTANCE	MOLECULAR WEIGHT	IONS	<i>i</i>	SODIUM CHLORIDE EQUIVALENT (E VALUE)
Antazoline phosphate	363	2	1.8	0.16
Antipyrine	188	1	1.0	0.17
Atropine sulfate-H ₂ O	695	3	2.6	0.12
Benoxinate hydrochloride	345	2	1.8	0.17
Benzalkonium chloride	360	2	1.8	0.16
Benzyl alcohol	108	1	1.0	0.30
Boric acid	61.8	1	1.0	0.52
Chloramphenicol	323	1	1.0	0.10
Chlorobutanol	177	1	1.0	0.24
Chlortetracycline hydrochloride	515	2	1.8	0.11
Cocaine hydrochloride	340	2	1.8	0.16
Cromolyn sodium	512	2	1.8	0.11
Cyclopentolate hydrochloride	328	2	1.8	0.18
Demecarium bromide	717	3	2.6	0.12
Dextrose (anhydrous)	180	1	1.0	0.18
Dextrose-H ₂ O	198	1	1.0	0.16
Dipivefrin hydrochloride	388	2	1.8	0.15
Ephedrine hydrochloride	202	2	1.8	0.29
Ephedrine sulfate	429	3	2.6	0.23
Epinephrine bitartrate	333	2	1.8	0.18
Epinephryl borate	209	1	1.0	0.16
Eucatropine hydrochloride	328	2	1.8	0.18
Fluorescein sodium	376	3	2.6	0.31
Glycerin	92	1	1.0	0.34
Homatropine hydrobromide	356	2	1.8	0.17
Hydroxyamphetamine hydrobromide	232	2	1.8	0.25
Idoxuridine	354	1	1.0	0.09
Lidocaine hydrochloride	289	2	1.8	0.22
Mannitol	182	1	1.0	0.18
Morphine sulfate-5H ₂ O	759	3	2.6	0.11
Naphazoline hydrochloride	247	2	1.8	0.27
Oxymetazoline hydrochloride	297	2	1.8	0.20
Oxytetracycline hydrochloride	497	2	1.8	0.12
Phenacaine hydrochloride	353	2	1.8	0.20
Phenobarbital sodium	254	2	1.8	0.24
Phenylephrine hydrochloride	204	2	1.8	0.32
Physostigmine salicylate	413	2	1.8	0.16
Physostigmine sulfate	649	3	2.6	0.13
Pilocarpine hydrochloride	245	2	1.8	0.24
Pilocarpine nitrate	271	2	1.8	0.23
Potassium biphosphate	136	2	1.8	0.43
Potassium chloride	74.5	2	1.8	0.76
Potassium iodide	166	2	1.8	0.34
Potassium nitrate	101	2	1.8	0.58
Potassium penicillin G	372	2	1.8	0.18
Procaine hydrochloride	273	2	1.8	0.21
Proparacaine hydrochloride	331	2	1.8	0.18
Scopolamine hydrobromide-3H ₂ O	438	2	1.8	0.12
Silver nitrate	170	2	1.8	0.33
Sodium bicarbonate	84	2	1.8	0.65
Sodium borate-10H ₂ O	381	5	4.2	0.42

Example Calculations of the Sodium Chloride Equivalent

- The sodium chloride equivalent of a substance may be calculated as follow

$$\frac{\text{Molecular weight of sodium chloride}}{\text{i Factor of sodium chloride}} \times \frac{\text{i factor of the substance}}{\text{Molecular weight of the substance}} = \text{Sodium chloride equivalent}$$

- Example-1.** Papaverine hydrochloride (m.w. 376) is a 2-ion electrolyte, dissociating 80% in a given concentration. Calculate its sodium chloride equivalent.
- Because papaverine hydrochloride is a 2-ion electrolyte, dissociating 80%, its i factor is 1.8.

Example-2. Calculate the sodium chloride equivalent for glycerin, a nonelectrolyte with a molecular weight of 92. Glycerin, i factor = 1.0.

Example-3. Calculate the sodium chloride equivalent for timolol maleate, which dissociates into two ions and has a molecular weight of 432.

Timolol maleate, i factor = 1.8.

- **Example 4.** Calculate the sodium chloride equivalent for fluorescein sodium, which dissociates into three ions and has a molecular weight of 376.

Fluorescein sodium, i factor = 2.6

Example Calculations of Tonicic Agent Required

1. How many grams of sodium chloride should be used in compounding the following prescription?

Rx Pilocarpine Nitrate 0.3g
Sodium Chloride q.s.
Purified Water ad 30ml
Make isoton. sol.
Sig. For the eye

Step 1. $0.23 \times 0.3 \text{ g} = 0.069 \text{ g}$ of sodium chloride represented by the pilocarpine nitrate

Step 2. $30 \times 0.009 = 0.270 \text{ g}$ of sodium chloride in 30 mL of an isotonic sodium chloride solution

Step 3. 0.270 g (from Step 2)

-0.069 g (from Step 1)

= **0.201 g of sodium chloride to be used.**

2. How many grams of boric acid should be used in compounding the following prescription?

R_x Phenacaine Hydrochloride 1%
Chlorobutanol 0.5%
Boric Acid q.S.
Purified Water ad 60ml
Make isoton. sol.
Sig. One drop in each eye.

- The prescription calls for 0.6 g of phenacaine hydrochloride and 0.3 g of chlorobutanol.

Step 1. $0.20 \times 0.6 \text{ g} = 0.120 \text{ g}$ of sodium chloride represented by phenacaine hydrochloride

$0.24 \times 0.3 \text{ g} = 0.072 \text{ g}$ of sodium chloride represented by chlorobutanol

Total: **0.192 g** of sodium chloride represented by both ingredients.

Step 2. $60 \times 0.009 = 0.540 \text{ g}$ of sodium chloride in 60 mL of an isotonic sodium chloride solution

Step 3. 0.540 g (from Step 2)

- 0.192 g (from Step 1)

= **0.348 g of sodium chloride** required to make the solution isotonic

But because the prescription calls for boric acid:

Step 4. $0.348 \text{ g} / 0.52$ (sodium chloride equivalent of boric acid) = **0.669 g of boric acid** to be used.

Using an Isotonic Sodium Chloride Solution to Prepare Other Isotonic Solutions

- A 0.9% w/v sodium chloride solution may be used to compound isotonic solutions of other drug substances as follows:

Step 1. Calculate the quantity of the drug substance needed to fill the prescription or medication order.

Step 2. Use the following equation to calculate the volume of water needed to render a solution of the drug substance isotonic:

$$\frac{\text{g of the drug} \times \text{drug's E value}}{0.009} = \text{mL of water needed to make an isotonic of the drug solution}$$

(the volume of the drug substance is considered negligible)

- **Step 3.** Add 0.9% w/v sodium chloride solution to complete the required volume of the prescription or medication order.

Example-1- *Using this method, determine the volume of purified water and 0.9% w/v sodium chloride solution needed to prepare 20 mL of a 1% w/v solution of hydromorphone hydrochloride ($E = 0.22$).*

Step 1. $20 \text{ mL} * 1\% \text{ w/v} = 0.2 \text{ g hydromorphone needed.}$

Step 2. $\frac{0.2 \text{ g} * 0.22}{0.009} = 4.89 \text{ mL purified water required to make an isotonic solution of hydromorphone hydrochloride.}$

Step 3. $20 \text{ mL} - 4.89 \text{ mL} = 15.11 \text{ mL } 0.9\% \text{ w/v sodium chloride solution required.}$

Proof: $20 \text{ mL} * 0.9\% = 0.18 \text{ g sodium chloride or equivalent required}$
 $0.2 * 0.22 = 0.044 \text{ g (sodium chloride represented by } 0.2 \text{ g hydromorphone hydrochloride).}$

$15.11 \text{ mL} * 0.9 \% = 0.136 \text{ g sodium chloride present}$
 $0.044 \text{ g} + 0.136 \text{ g} = 0.18 \text{ g sodium chloride required for isotonicity}$

Use of Freezing Point Data in Isotonicity Calculations

- Freezing point data (T_f) can be used in isotonicity calculations when the agent has a tonicic effect and does not penetrate the biologic membranes in question (e.g., red blood cells).
- As stated previously, the freezing point of both blood and lacrimal fluid is 0.52C.
- Thus, a pharmaceutical solution that has a freezing point of 0.52C is considered isotonic.

Freezing Point Isotonicity Cont'd

- Freezing point depression data may be used in isotonicity calculations as shown by the following.

Table: FREEZING POINT DATA FOR SELECT AGENTS

AGENT	FREEZING POINT DEPRESSION, 1% SOLUTIONS (ΔT_f°)
Atropine sulfate	0.07
Boric acid	0.29
Butacaine sulfate	0.12
Chloramphenicol	0.06
Chlorobutanol	0.14
Dextrose	0.09
Dibucaine hydrochloride	0.08
Ephedrine sulfate	0.13
Epinephrine bitartrate	0.10
Ethylmorphine hydrochloride	0.09
Glycerin	0.20
Homatropine hydrobromide	0.11
Lidocaine hydrochloride	0.063
Lincomycin	0.09
Morphine sulfate	0.08
Naphazoline hydrochloride	0.16
Physostigmine salicylate	0.09
Pilocarpine nitrate	0.14
Sodium bisulfite	0.36
Sodium chloride	0.58
Sulfacetamide sodium	0.14
Zinc sulfate	0.09

Example Calculations Using Freezing Point Data

Question 1. *How many milligrams each of sodium chloride and dibucaine hydrochloride are required to prepare 30 mL of a 1% solution of dibucaine hydrochloride isotonic with tears?*

Buffers and Buffer Solution

- The presence of certain substances or combinations of substances in aqueous solution imparts to the system **the ability to maintain a desired pH at a relatively constant level**, even with the addition of materials that may be expected to change the hydrogen–ion concentration.
- These substances or combinations of substances are called ***buffers***;
- **Buffers** ability to resist changes in pH is referred to as *buffer action*; their efficiency is measured by the function known as *buffer capacity*; and solutions of them are called ***buffer solutions***.
- By definition, then, a ***buffer solution*** is a system, usually an aqueous solution, that possesses the **property of resisting changes in pH** with the addition of small amounts of a strong acid or base.

Buffers Cont'd

- Buffers are used to establish and maintain an ion activity within rather narrow limits.
- **In pharmacy, the most common buffer systems are used in**
- (i) the preparation of such dosage forms injections and ophthalmic solutions, which are placed directly into pH sensitive body fluids;
- (ii) the manufacture of formulations in which the pH must be maintained at a relatively constant level to ensure maximum product stability; and
- (iii) pharmaceutical tests and assays requiring adjustment to or maintenance of a specific pH for analytic purposes.

Buffers Cont'd

- A buffer solution is usually composed of a weak acid and a salt of the acid, such as acetic acid and sodium acetate, or a weak base and a salt of the base, such as ammonium hydroxide and ammonium chloride.
- Typical buffer systems that may be used in pharmaceutical formulations include the following pairs:
acetic acid and sodium acetate, boric acid and sodium borate, and disodium phosphate and sodium acid phosphate.

Buffers Cont'd

- In the selection of a buffer system, due consideration must be given to the dissociation constant of the weak acid or base to ensure maximum buffer capacity.
- This dissociation constant, in the case of an acid, is a measure of the strength of the acid; the more readily the acid dissociates, the higher its dissociation constant and the stronger the acid.
- Selected dissociation constants, or K_a values, are given in the next Table.

Buffers Cont'd

Table: DISSOCIATION CONSTANTS OF SOME WEAK ACIDS At 25C

ACID	K_a
Acetic	1.75×10^{-5}
Barbituric	1.05×10^{-4}
Benzoic	6.30×10^{-5}
Boric	6.4×10^{-10}
Formic	1.76×10^{-4}
Lactic	1.38×10^{-4}
Mandelic	4.29×10^{-4}
Salicylic	1.06×10^{-3}

Buffers Cont'd

- The dissociation constant, or K_a value, of a weak acid is given by the equation:

$$K_a = \frac{(H^+) (A^-)}{(HA)} \quad \begin{array}{l} \text{where } A^- = \text{salt} \\ HA = \text{acid} \end{array}$$

- Because the numeric values of most dissociation constants are small numbers and may vary over many powers of 10, it is more convenient to express them as negative logarithms:

$$\mathbf{pK_a = -\log K_a}$$

Buffers Cont'd

When equation $K_a = \frac{(H^+) (A^-)}{(HA)}$ is expressed in logarithmic form, it is written:

$$pK_a = -\log (H^+) - \log \frac{\text{salt}}{\text{acid}}$$

and because $pH = -\log (H^+)$:

then

$$pK_a = pH - \log \frac{\text{salt}}{\text{acid}}$$

and

$$pH = pK_a + \log \frac{\text{salt}}{\text{acid}}$$

Buffers Cont'd

Similarly, the dissociation constant, or K_b value, of a weak base is given by the equation:

$$K_b = \frac{(B^+) (OH^-)}{(BOH)} \quad \begin{array}{l} \text{in which } B^+ = \text{salt} \\ \text{and } BOH = \text{base} \end{array}$$

and the buffer equation for weak bases, which is derived from this relationship, may be expressed as:

$$pH = pK_w - pK_b + \log \frac{\text{base}}{\text{salt}}$$

Questions

1. Calculating the pKa value of a weak acid, given its dissociation constant, K_a : *The dissociation constant of acetic acid is 1.75×10^{-5} at 25°C. Calculate its pKa value.*
2. *What is the pH of a buffer solution prepared with 0.05 M sodium borate and 0.005 M boric acid? The pKa value of boric acid is 9.24 at 25°C.*
3. *What is the pH of a buffer solution prepared with 0.05 M ammonia and 0.05 M ammonium chloride? The K_b value of ammonia is 1.80×10^{-5} at 25°C.*